Information-gain AUC in development worlds

Qwen3-32B · pooled polarity-balanced axis · extraction indices 0–49; development indices 50–99

All four layers and seven positions retained; axis sign and classification threshold are unchanged from extraction.

Objects · New-information question

Layer index	Record	Prompt end	Token 32	Token 128	Token 512	Think end	Answer boundary
15	0.907 n=50/50	0.899 n=50/50	0.578 n=50/50	0.564 n=49/50	NA no axis n=0/50	0.863 n=50/50	0.770 n=50/50
31	1.000 n=50/50	0.989 n=50/50	0.586 n=50/50	0.598 n=49/50	NA no axis n=0/50	0.833 n=50/50	0.853 n=50/50
47	1.000 n=50/50	0.992 n=50/50	0.608 n=50/50	0.612 n=49/50	NA no axis n=0/50	0.797 n=50/50	0.853 n=50/50
57	1.000 n=50/50	0.982 n=50/50	0.577 n=50/50	0.597 n=49/50	NA no axis n=0/50	0.891 n=50/50	1.000 n=50/50

Objects · Already-known question

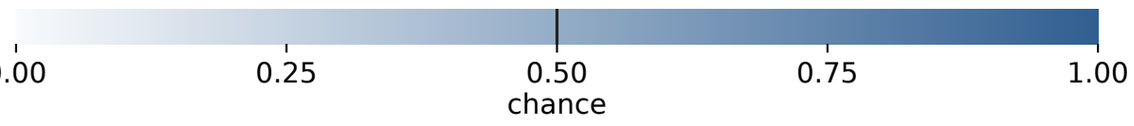
Layer index	Record	Prompt end	Token 32	Token 128	Token 512	Think end	Answer boundary
15	0.918 n=50/50	0.804 n=50/50	0.564 n=50/50	0.653 n=50/50	NA no axis n=0/50	0.819 n=50/50	0.754 n=50/50
31	1.000 n=50/50	0.987 n=50/50	0.622 n=50/50	0.712 n=50/50	NA no axis n=0/50	0.813 n=50/50	0.848 n=50/50
47	1.000 n=50/50	0.995 n=50/50	0.607 n=50/50	0.713 n=50/50	NA no axis n=0/50	0.742 n=50/50	0.858 n=50/50
57	1.000 n=50/50	0.993 n=50/50	0.627 n=50/50	0.651 n=50/50	NA no axis n=0/50	0.904 n=50/50	1.000 n=50/50

Machines · New-information question

Layer index	Record	Prompt end	Token 32	Token 128	Token 512	Think end	Answer boundary
15	0.907 n=50/50	0.926 n=50/50	0.640 n=50/50	0.690 n=50/50	NA no axis n=0/50	0.890 n=50/50	0.885 n=50/50
31	0.996 n=50/50	0.993 n=50/50	0.672 n=50/50	0.738 n=50/50	NA no axis n=0/50	0.927 n=50/50	0.948 n=50/50
47	1.000 n=50/50	0.995 n=50/50	0.679 n=50/50	0.748 n=50/50	NA no axis n=0/50	0.887 n=50/50	0.957 n=50/50
57	0.989 n=50/50	0.994 n=50/50	0.681 n=50/50	0.778 n=50/50	NA no axis n=0/50	0.965 n=50/50	1.000 n=50/50

Machines · Already-known question

Layer index	Record	Prompt end	Token 32	Token 128	Token 512	Think end	Answer boundary
15	0.912 n=50/50	0.889 n=50/50	0.666 n=50/50	0.532 n=50/50	NA no axis n=0/50	0.896 n=50/50	0.854 n=50/50
31	0.997 n=50/50	0.992 n=50/50	0.695 n=50/50	0.588 n=50/50	NA no axis n=0/50	0.956 n=50/50	0.976 n=50/50
47	1.000 n=50/50	0.996 n=50/50	0.684 n=50/50	0.595 n=50/50	NA no axis n=0/50	0.900 n=50/50	0.973 n=50/50
57	0.993 n=50/50	0.992 n=50/50	0.724 n=50/50	0.639 n=50/50	NA no axis n=0/50	0.986 n=50/50	1.000 n=50/50



Development only. Original whole-instrument gate: not met (count validity). Semantic-world overlap: objects 19; machines 0.

There are 82 exact-prompt overlaps between extraction and development. This is not a fresh confirmation sample.

Record precedes the question: its context is identical across task kinds, so cross-kind decoding is not independent replication.

n = complete scored four-cell worlds / 50 assigned. No axis = no complete paired extraction worlds. Available world sets can differ by position.

These are descriptive projections. They establish no causal mechanism, motivational state, or subjective experience.